

Bit/Symbol Synchronization Update

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Summary

The bit/symbol synchronization in `sdr_ch_t` navigation decoding was changed from an absolute-amplitude threshold on a single epoch to the bit-energy maximum-likelihood (ML) method of Kokkonen and Pietila. The goal is to make the synchronized symbol boundary (and therefore `ch->tow`) deterministic, addressing GitHub issue #83.

The change is contained entirely in `sync_symb()` in `src/sdr_nav.c`. No data structures, initialization, or per-signal decoders were modified.

Problem (issue #83)

`sync_symb()` declared symbol sync on the first epoch where:

1. the last 200 ms of IP correlations split into N-ms groups all had a uniform sign (`R[i] == +/-N`),
2. the two most recent groups had opposite sign, and
3. the mean IP magnitude of the latest group exceeded `THRES_SYNC = 0.05`.

`THRES_SYNC` is an absolute amplitude. Its effective value scales with front-end bit width, AGC, and signal level, so it is not a fixed operating point:

- At strong signal (around C/N0 50 dB-Hz) the amplitude gate passes trivially and no longer suppresses residual pull-in transients. A half-cycle-slip or PLL pull-in transient left in the IP history can mimic a bit transition at an arbitrary 1 ms position, so the declared boundary can shift by +/-1 ms between runs. For L1C/A this is a silent error: 19 of 20 samples still carry the correct sign, so frame sync and parity still pass while `ch->tow` is off by 1 ms, producing a pseudorange error.
- At weak signal the same fixed threshold can block sync entirely.

Steady-state symbol errors do not explain the strong-signal randomness; at 50 dB-Hz the 1 ms coherent SNR is high and the sign sequence is effectively error-free. The randomness comes from transients in the evaluation window, not from the data.

The *silent* failure mode is specific to the 20 ms cases (L1C/A, NavIC I5S): a 1 ms boundary error leaves 19 of 20 samples correct, so frame sync and parity still pass while `ch->tow` is wrong. The 2 ms cases (SBAS, BDS B1I D2) instead self-recover through downstream FEC/CRC, because a 1 ms boundary error destroys the symbol stream rather than shifting it silently. The replacement sync itself is general and applies to all `sync_symb()` callers (see Properties).

Method

`sync_sync()` now uses the bit-energy ML method (Kokkonen and Pietila, IEEE PLANS 2002, pp. 85-90; see also Van Dierendonck, ch. 8 in Parkinson and Spilker, *Global Positioning System: Theory and Applications*, Vol. I, AIAA, 1996 for the classical histogram method this replaces).

For each of the N possible boundary phases $m = 0..N-1$, both the coherent and the incoherent N -ms symbol energy are accumulated over K symbols:

$$E[m] = \text{sum over } k \text{ of } \left| \text{sum of IP over the } N \text{ samples of symbol } k \text{ at phase } m \right|$$

$$A[m] = \text{sum over } k \text{ of } \text{sum of } |IP| \text{ over the same } N \text{ samples}$$

$E[m]/A[m]$ is the coherent purity: 1.0 when every symbol at phase m is sign-consistent, lower as the phase straddles bit transitions. The boundary phase is

$$m0 = \text{argmax}_m (E[m] / A[m])$$

chosen by purity rather than raw energy, so the choice is scale-invariant even between phases with unequal $|IP|$ sums. Sync is declared only when all three of the following hold:

$\text{coh} = E[m0] / A[m0]$	$\geq 0.5 (1 + 1/\sqrt{N})$	(a)
$\text{margin} = \text{coh} - \max(\text{coh}[m0-1], \text{coh}[m0+1])$	$\geq \text{THRES_SYNC_M}$	(b)
$E[m0] \geq \text{THRES_SYNC_R} * E[(m0 + N/2) \% N]$		(c)

(a) requires real coherent structure (rejects noise, whose purity is $\sim 1/\sqrt{N}$). (c), the opposite-phase energy ratio, rejects the coarse half-symbol ($N/2$) error: a half-bit shift fully cancels the coherent sum on a transition, so the clean-signal ratio is ~ 2.0 for any N and $\text{THRES_SYNC_R} = 1.4$ rejects the no-peak (ratio ~ 1) case. (c) does **not** by itself separate the ± 1 ms neighbours that are the core of issue #83, because an adjacent phase still keeps $(N-1)/N$ of the coherent sum. (b), the adjacent-phase coherence margin, is the gate for that 1 ms ambiguity: $m0$ must beat *both* neighbours in purity by THRES_SYNC_M . When the top two phases are within the margin (a lone transition, a near-DC window, or noise at low $C/N0$) no boundary is declared; the channel keeps accumulating until K grows enough to separate them, which preserves determinism instead of committing to a 1 ms-ambiguous estimate.

Because $E[]$ and $A[]$ are deterministic functions of the IP history, the same input produces the same $m0$. A single transient sign flip perturbs only the K windows that contain it; spread over K accumulated symbols its effect is diluted relative to a first-hit test, so it is unlikely to create a spurious peak or shift the boundary. (The loss is not exactly uniform across phases — it depends on where the outlier falls relative to each phase's windows.)

Properties

- Scale-invariant: phase selection uses the per-phase purity E/A , and the gates use ratios/margins, never an absolute amplitude.
- Deterministic: argmax over accumulated coherent purity (E/A) plus an explicit confidence margin, so a boundary is committed only when the estimate is unambiguous.
- Soft-decision and bit-energy based, so it also synchronizes at lower C/N0 than the old "all-200-ms-samples-uniform-sign" requirement.
- General across signals (intended, not a side effect): the same scale-invariant rule serves every `sync_symb()` caller — L1C/A and I5S (N=20), G1CA (N=10), SBAS and B1I D2 (N=2). Unit tests in `test/utest` cover N=2, 10, 20; receiver-level behaviour is verified on GLONASS G1CA (N=10) playback. The half-symbol contrast (c) is 2:1 for every N; the 1 ms margin (b) is only meaningful for N>2 (for N=2 the adjacent and opposite phases coincide).

Known limitation

`THRES_SYNC_M = 0.03` is a conservative initial value. For random L1C/A data the structural purity gap between the true phase and its 1 ms neighbour is only about 0.05 ($20/20$ vs $19/20$), so the margin must stay well below that to avoid blocking valid sync, yet be large enough to reject noise-driven ties — a narrow window that has not yet been tuned against real low-C/N0 IF data. Signals with a dedicated meander (G1CA, where every 10 ms boundary transitions) have a much larger gap (~0.2) and are insensitive to the exact value. If weak channels are seen to stall at symbol sync, lower `THRES_SYNC_M`; if 1 ms boundary jitter persists at strong signal, raise it.

Transient guard

The accumulation window is floored at lock count `Lmin = T_SYNC_START / ch->T`, with `T_SYNC_START = 1.5 s` (equal to `T_NPULLIN` in `src/sdr_ch.c`, where nav decoding begins). The oldest IP sample used is guaranteed to be at or after `Lmin`, which is after the PLL pull-in (`T_FPULLIN = 1.0 s`). This excludes the 1.0-1.5 s pull-in transient that was the root cause of the strong-signal randomness.

Symbol-lost detection

The periodic branch keeps the original lenient behavior: symbol lost only when the mean IP magnitude falls below an absolute `THRES_LOST = 0.001`. The real loss-of-lock is handled by the C/N0 unlock gate in `src/sdr_ch.c`, so this is a backstop that almost never fires while a channel tracks.

An earlier revision of this change replaced it with a scale-invariant per-symbol coherent-vs-incoherent ratio (`|mean_IP(N)| >= 0.5 * mean(|IP|)`). That regressed low-C/N0 frame lock: a single N-ms symbol has high variance, so at ~30 dB-Hz the ratio dipped below the threshold most symbol periods, resetting `ssync` every ~N ms. The channel then produced almost no symbols (it cycled SYMBOL SYNC / SYMBOL LOST instead of adding to the symbol buffer) and frame sync never completed. Verified on a GLONASS G1CA (N = 10) playback: strong channels (~49 dB-Hz) were unaffected, weak channels (30-35 dB-Hz) lost frame lock. The per-symbol ratio test was reverted; the deterministic K-P sync (the actual fix

for issue #83) is unchanged. A scale-invariant lost detector would have to accumulate over several symbols (or use hysteresis) to avoid this thrashing.

Parameters

Defined at the top of `src/sdr_nav.c`:

Name	Value	Meaning
<code>SYNC_NMAX</code>	20	max symbol length (ms); sizes the phase and prefix-sum arrays
<code>K_SYNC_MIN</code>	20	min symbols accumulated before sync may be declared
<code>K_SYNC_MAX</code>	100	max symbols accumulated (caps lookback and stack use)
<code>THRES_SYNC_R</code>	1.4	sync energy ratio, peak vs opposite (half-symbol) phase
<code>THRES_SYNC_M</code>	0.03	min coherence margin of the peak over its 1 ms neighbours
<code>THRES_LOST</code>	0.001	symbol-lost absolute threshold (kept from original)
<code>T_SYNC_START</code>	1.5	accumulation history floor (s), equals <code>T_NPULLIN</code>

Removed: the old `THRES_SYNC = 0.05` absolute sync threshold in `src/sdr_nav.c`, replaced by the K-P energy ratio. `THRES_LOST = 0.001` is retained for the symbol-lost backstop. (The identically named macros in `src/sdr_ch.c` for secondary-code sync are unrelated and unchanged.)

Implementation notes

- The K-P metric is computed statelessly from the existing `ch->trk->P[]` history (`SDR_N_HIST = 5000` deep, i.e. 250 symbols at $N = 20$), so no new `sdr_nav_t` field is required.
- Window sums use two prefix-sum arrays over the lookback range (one for IP, one for $|IP|$, giving `E[m]` and `A[m]`), reducing the cost from $O(N^2 K)$ to $O(N K)$ per epoch. The heavy branch runs only while a channel is unsynced (`ssync == 0`), a short startup transient.
- The boundary is stored as `ch->nav->ssync = ch->lock - m0 - N`, the same convention as the old code (the old value `ch->lock - N` is the `m0 = 0` case). Only `ssync mod N` and `ssync != 0` are significant for the five `sync_symb()` callers; the absolute `ssync` timestamp used by the `+1104` and `+6680` checks belongs to the separate `sync_sec_code()` path (G1OCD, G3OCD) and is unaffected.

Trade-offs

- Time to first symbol sync increases by at least `K_SYNC_MIN` symbols, and further if the adjacent-phase margin needs more accumulation to clear `THRES_SYNC_M` (the window grows with lock up to `K_SYNC_MAX`). For L1C/A this is roughly 0.3-0.5 s later than before at strong signal, more at low C/N_0 .
- Parameter values (`THRES_SYNC_R = 1.4`, `THRES_SYNC_M = 0.03`, `K_SYNC_MIN = 20`, `K_SYNC_MAX = 100`) are analysis-based initial settings. They should be reviewed against the issue

#83 reproduction data, in particular the observed distributions of the opposite-phase energy ratio and the adjacent-phase margin at high and low C/N0.

Verification

Unit tests (MSYS2 UCRT64, the project toolchain):

```
make -C test/utest test          # full suite, or:  
make -C test/utest test_sdr_nav  # nav decoder only, then ./test_sdr_nav.exe
```

`test_sdr_nav` covers symbol sync for N = 20 (L1C/A), 10 (G1CA) and 2 (SBAS), the adjacent-margin rejection of a flat (transition-free) window, and the symbol-lost backstop.

The `$LOG ... SYMBOL SYNC (ratio, coh, margin)` message reports the peak-to- opposite energy ratio (capped at 99.99 when the opposite phase has no energy), the coherent purity `coh`, and the adjacent-phase `margin`. Receiver-level acceptance against the issue #83 reproduction is still the deciding test: across repeated runs of the same IF data, `ssync mod N` must be identical at strong signal, the three logged values stable, and weak channels must still reach sync. The same run is used to tune `THRES_SYNC_M` and `THRES_SYNC_R`.